4. To remove each entry, use:

```
setfacl -x "entry"
```

5. To display permissions, use:

```
getfacl filename
```

—Sandhya Bankar, bankarsandhya512@gmail.com

Get your IP address in the command line

If you ever need the public IP address of your computer, you can get it from the command line by adding one simple alias statement to the *.bashrc* file located in the home directory.

Add the alias as follows:

```
myip='echo $(curl --silent ipecho.net/plain)'
```

… at the end of *.bashrc* and reopen the terminal; or you can run sources, as follows:

```
~/.bashrc
```

The next time you need a public IP address, there's no need to visit the browser—just type the following command:

```
#myip
```

Your computer's public IP address will be displayed.

An example is:

```
$user@linux]$ myip
```

The output will be your public IP, i.e.,

```
1xx.2xx.0.58
```

—Ankith Herle, ankitherle@gmail.com

Start/stop services automatically on booting up

To start/stop a service when booting up is easy using the startup manager tool, but this doesn't show the status of all the services on bootup—whether they start automatically or not. To change this or view the status of all services on booting up, you can use the *systemctl* utility on Linux systems that have *systemd* as the *init* manager. Here are a few generic commands and what they do.

To see all services available in your Linux system, use the following command:

```
$systemctl  | grep service
```

To see all active services, use:

```
$systemctl list-units
```

To start or stop a service on booting up, use:

```
$sudo systemctl enable application.service
$sudo systemctl disable application.service
```

To see the status of the service, use:

```
$systemctl status application.service
```

For this, you can also try the following command:

```
$systemctl is-enabled application.service
```

—Shubham Dubey, sdubey504@gmail.com

Base conversion using shell

We often need to convert a number from one base to another—for instance, convert from decimal to hexadecimal or from octal to hexadecimal, and so on. There are many GUI applications available for this; however, we can do this very quickly and efficiently from the CLI as well. It simply uses shell's built-in 'printf' command. Let us understand it with simple examples.

The prototype of the 'printf' function is:

```
printf FORMAT [ARGUMENT] ...
```

It uses the '%o' format specifier for octal (base 8) numbers, '%f' for hexadecimal (base 16) numbers and '%d' or '%i' for decimal (base 10) numbers. Shown below are simple examples that illustrate conversions:

```
[bash]$ printf "DEC(%i) = HEX(%X)\n" 255 255   # decimal to
hexadecimal conversion
DEC(255) = HEX(FF)

[bash]$ printf "HEX(%X) = OCT(%o)\n" 0xFF 0xFF # hexadecimal
to octal conversion
HEX(FF) = OCT(377)

[bash]$ printf "OCT(%o) = DEC(%i)\n" 255 255   # octal to
decimal conversion
OCT(377) = DEC(255)
```

—Narendra Kangralkar, narendrakangralkar@gmail.com

END